

Atul Gupta

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Summary

Dynamic AIML Engineer with 5+ years of experience in artificial intelligence, machine learning, data modeling, and cloud engineering. 3x AWS certified.

Technical Skills

Languages: Python, SQL, PowerShell

Backend & APIs: FastAPI, Flask, Pydantic, REST APIs, JSend, python-pptx

AI/ML: Azure OpenAI, Agno, LangChain, PyTorch, scikit-learn, XGBoost, TensorFlow, prompt engineering, LLM guardrails

Data & Databases: pandas, NumPy, Databricks SQL, SQL Server, MongoDB, Oracle, DynamoDB, AWS Glue, Airflow

Cloud & DevOps: AWS, SageMaker, Bedrock, Lambda, CloudWatch, SNS, S3, Docker, Jenkins

Tools & Reporting: Tableau, SSRS, GitHub, Jira, ServiceNow, Jupyter Notebook

Experience

Specialist Programmer (Full Stack AI Engineer for Qualcomm)

Aug 2025 – Present

Infosys Limited

- Engineered backend and data services for **Slide Studio**, an AI-powered PowerPoint automation platform that generates finance decks from reusable PPT templates, slide structure, Databricks-backed data, and LLM-generated transformer logic.
- Built a **human-in-the-loop AI code generation workflow** where LLMs combined PPT structure, financial datasets, user-provided formulas, and business transformation rules to generate custom Python transformer code for automated slide population.
- Designed an AI prompt-optimization layer that converted vague user intent into structured, reviewable prompts, allowing users to refine finance-specific formulas, mappings, and transformation logic before code generation.
- Implemented a secure guardrails pipeline for **LLM-generated transformer scripts**, validating code before disk persistence or runtime execution to prevent unsafe operations and protect backend systems.
- Developed FastAPI services for **data retrieval, prompt generation, transformer generation, guardrail validation, transformer execution, template population, AI commentary, and PPT export** using **Python, Databricks SQL, MongoDB, Azure OpenAI GPT-4o, and python-pptx**.
- Built a dynamic PowerPoint rendering engine using **python-pptx**, supporting reusable templates, structured slide layouts, table population, placeholder replacement, formatting, slide merging, and deck export.
- Delivered the **Report360 Unscheduled PO / Push Out analytics backend**, identifying placed orders without assigned shipping schedules and surfacing revenue-timing risks across customer, product, business unit, and fiscal dimensions.
- Built a dedicated **Python/pandas analytics toolkit** to compute deeper insights beyond the UI-visible report grid, including customer-level prioritization, trend analysis, metric rollups, and business-driver summaries.
- Integrated the analytics toolkit into an **Agno-based LLM tool workflow**, enabling asynchronous AI commentary that reasoned over computed backend insights to explain the situation, identify key drivers, and recommend customer focus areas.
- Built 10 cache-first FastAPI lookup APIs for finance workflows, using MongoDB with 12-hour TTL caching and Databricks SQL fallback to serve fresh ledger, period, account, department, product, business-line, and currency-code reference data through standardized JSend responses.

Application Development Analyst

Jan 2019 – Jul 2022

Accenture

- Built AWS-based ETL and ML pipelines using **Python, SQL Server, AWS Glue, Airflow, Docker, and SageMaker** to process Hilton reservation, transaction, and guest behavior data for forecasting and churn analytics.
- Developed a COVID-era revenue forecasting model for Hilton's top 50 properties using **XGBoost/PyTorch**, achieving **19% MAPE** and integrating outputs into **SSRS** daily executive reports.
- Built a Hilton Honors churn prediction model using 2 years of loyalty data, identifying churn-prone members with **85% accuracy** to support targeted retention campaigns.
- Automated patch deployment and monitoring across Hilton's global OnQ PMS infrastructure using **Python, PowerShell, ServiceNow, AWS Lambda, CloudWatch, and SNS**, reducing manual effort by **80%**, downtime by **90%**, and deployment failures by **90%**.
- Managed **6,000+ SQL Server production database instances** and supported enterprise incident/change management, resolving **150+ critical issues** and maintaining compliance-driven stored procedures and jobs.
- Led **15 automation initiatives**, including **7 Six Sigma projects**, saving **10,000 annual man-hours** and **\$30K** in third-party service costs; recognized as **Accenture TechStar**.

Education

Master of Science in Computer Science (GPA: 3.96): Stevens Institute of Technology, Hoboken, NJ

Sep 2022 – May 2024

- Courses:** Machine Learning, Deep Learning, Data Mining with R, Big Data, Agile development, Algorithms, Statistics
- Tech:** Sklearn, PySpark, MongoDB, Java, Python, Tensorflow, Keras

Bachelor of Technology in Electrical and Electronics Engineering: VIT University

Jun 2013 – Jun 2017

Certifications

AWS Certified Machine Learning Engineer Associate ([Link](#))

AWS Certified Solutions Architect Associate ([Link](#))

AWS Certified AI Practitioner ([Link](#))